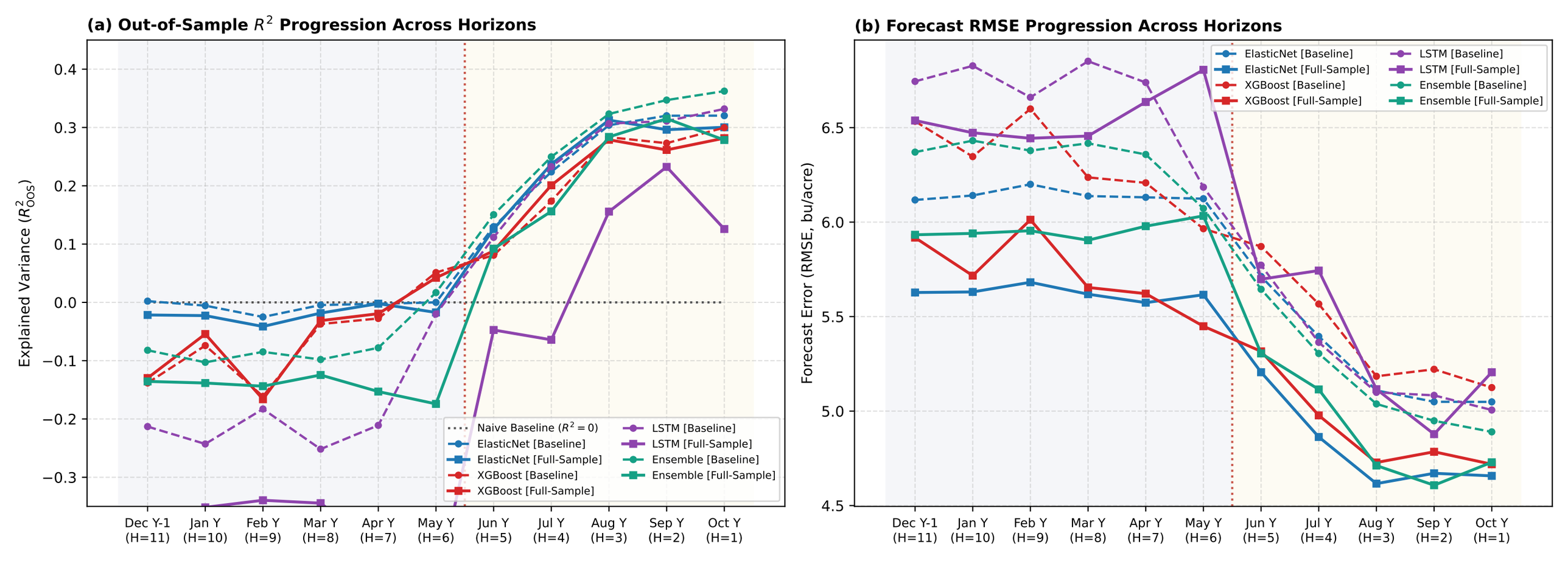
(a) Out-of-Sample R^2 Progression Across Horizons
Explained Variance (R^2_{OOS})
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
Dec Y-1 (H=11)
Jan Y (H=10)
Feb Y (H=9)
Mar Y (H=8)
Apr Y (H=7)
May Y (H=6)
Jun Y (H=5)
Jul Y (H=4)
Aug Y (H=3)
Sep Y (H=2)
Oct Y (H=1)
Naive Baseline ($R^2 = 0$)
ElasticNet [Baseline]
ElasticNet [Full-Sample]
XGBoost [Baseline]
XGBoost [Full-Sample]
LSTM [Baseline]
LSTM [Full-Sample]
Ensemble [Baseline]
Ensemble [Full-Sample]
(b) Forecast RMSE Progression Across Horizons
Forecast Error (RMSE, bu/acre)
6.5
6.0
5.5
5.0
4.5
Dec Y-1 (H=11)
Jan Y (H=10)
Feb Y (H=9)
Mar Y (H=8)
Apr Y (H=7)
May Y (H=6)
Jun Y (H=5)
Jul Y (H=4)
Aug Y (H=3)
Sep Y (H=2)
Oct Y (H=1)
ElasticNet [Baseline]
ElasticNet [Full-Sample]
XGBoost [Baseline]
XGBoost [Full-Sample]
LSTM [Baseline]
LSTM [Full-Sample]
Ensemble [Baseline]
Ensemble [Full-Sample]